

Instructions

Requirements according to the work order

Must

- The database can:
 - Save the users with their first name, last name, email and password
 - Save the different nodes with their name, the owner and the date on which it was assigned to the plant species
 - Save plant species
 - Save the measurements
 - Save the tasks
- The documentation contains the MCD and the MLD

User A user (eng. Operator) is a person who can log in to the system. The login to the website is done by e-mail in combination with a password.

A user also has a first name & last name.

Nodes A node is responsible for a maximum of one plant species. Each node can have a name. If a node is planted, it contains the date on which it was assigned to the plant species. Each node has an owner. The default owner of a node is the user "admin@ecample.com" (id=1). For unique identification, each node has a MAC address.

Plant species A plant species (eng. Variety) is defined by its common name. A botanical name can also be given. To optimally supply plants with light and water, we store the optimal threshold values for brightness and soil moisture for each plant species. Guideline values for air temperature and humidity can also be stored.

Measurements A measurement contains various sensor data that captures the following values:

- Soil moisture (indicates the moisture in the soil)
- Air temperature (detects the temperature of the environment)
- Humidity (measures the water vapor content in the air)
- Brightness (lumens) (indicates light intensity)
- State of the light (shows in percentage how much the node's grow light is set)

In addition, the timestamp is recorded, which indicates the exact time of the measurement, as well as the node affiliation, which indicates which node the data comes from. If a measurement cannot be read or is outside a reasonable range, it is stored as NULL.

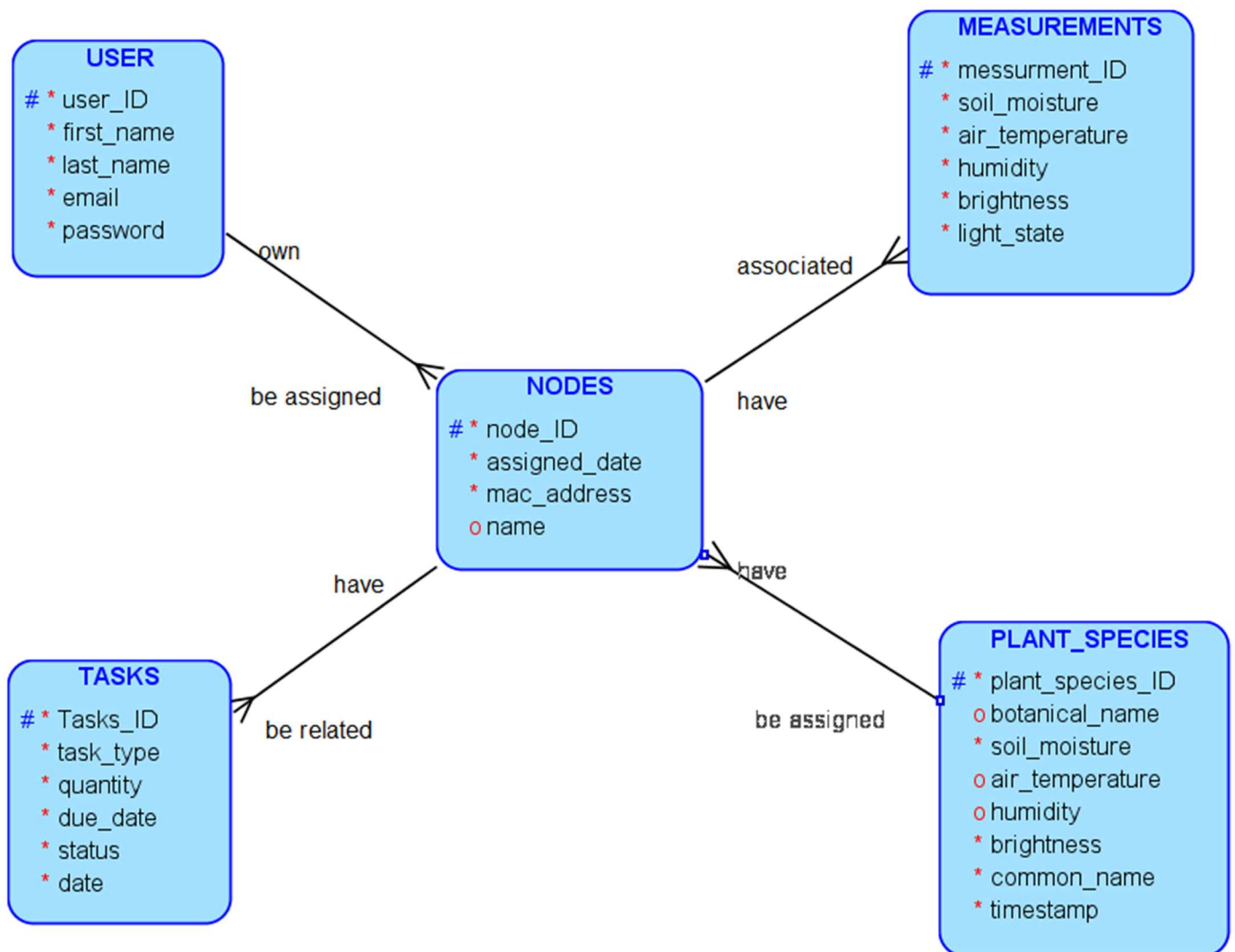
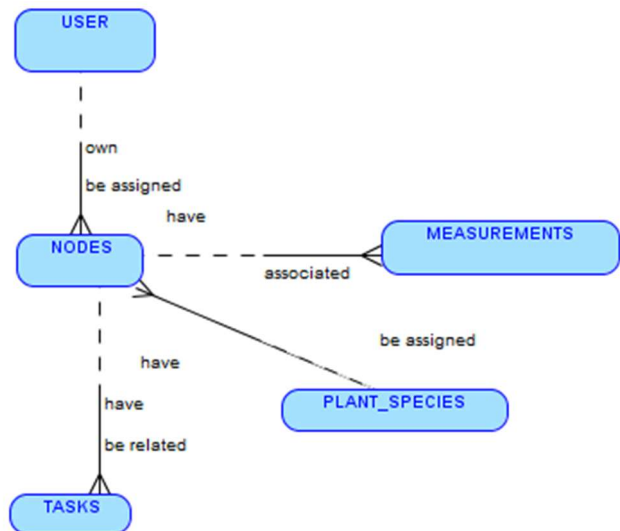
Tasks The system & users give up tasks that the nodes must do. A task corresponds to either lighting or watering. Values between 0-500 can be specified as quantities. Each task has a due date and time, which is "now" by default. New tasks are always "not done" by default. A task is marked as complete once it is completed.

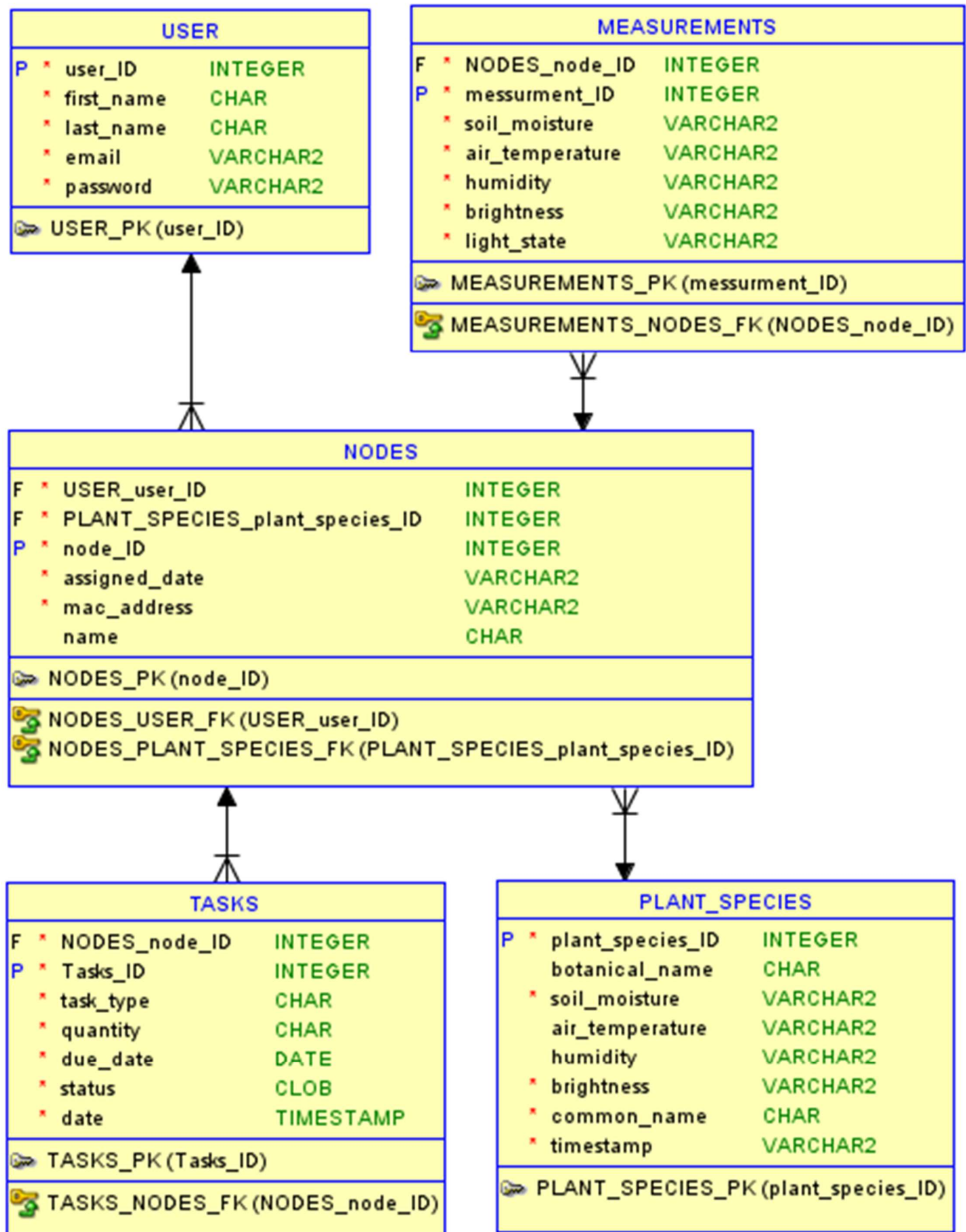
Each USER must own one or more NODES. Each NODES must be assigned to one and only one USER.

Each PLANT-SPECIES must be assigned to one or more NODES. Each NODES must be assigned to one and only one PLANT.

Each NODES must have one or more MEASUREMENTS. Each MEASUREMENTS must be associated with one and only one NODES.

Each TASKS must be related to one and only one NODES. Each NODES must have one or more TASKS.





- Mapping the relationships

USER				MESSURMENT			
Key-Types	Optimality	Column-name	Type	Key-Types	Optimality	Column-name	Type
PK	*	user_ID	INTERGER	FK	*	node_ID	INTERGER
	*	first_name	VARCHAR2	PK	*	messurment_ID	INTERGER
	*	last_name	VARCHAR2		*	soil_moisture	number
	*	email	VARCHAR2		*	air_temperature	number
	*	password	VARCHAR2		*	humidity	number
FK	*	node_ID	INTERGER		*	brightness	number
					*	light_state	number

NODES				TASKS			
Key-Types	Optimality	Column-name	Type	Key-Types	Optimality	Column-name	Type
FK	*	user_ID	INTERGER	FK	*	node_ID	INTERGER
FK	*	plant_species_ID	INTERGER	PK	*	tasks_ID	INTERGER
PK	*	node_ID	INTERGER		*	task_type	VARCHAR2
	*	assigned_date	DATE		*	quantity	number
	*	mac_address_	VARCHAR2		*	due_date	DATE
	°	name	VARCHAR2		*	status	VARCHAR2
					*	date	DATE

PLANT_SPECIES			
Key-Types	Optimality	Column-name	Type
PK	*	plant_species_ID	INTERGER
		botanic_name	VARCHAR2
	*	soil_moisture	number
		air_temperature	number
		humidity	number
	*	brightness	number
	*	common_name	CHAR
	*	timestamp	TIMESTAMP

- Create the matrix diagram

	USER	NODES	PLANT-SPECIES	MEASUREMENTS	TASKS
USER		own			
NODES	assigned		assigned	have	have
PLANT-SPECIES		assigned			
MEASUREMENTS		associated			
TASKS		related			

- For each relationship write the sentences

Each USER must own one or more NODES. Each NODES must be assigned to one and only one USER.

Each PLANT-SPECIES must be assigned to one or more NODES. Each NODES must be assigned to one and only one PLANT.

Each NODES must have one or more MEASUREMENTS. Each MEASUREMENTS must be associated with one and only one NODES.

Each TASKS must be related to one and only one NODES. Each NODES must have one or more TASKS.

- Using Apex, create the table for each entity

USER

Column Name	Data Type
USER_ID	NUMBER(6,0)
FIRST_NAME	VARCHAR2(40)
LAST_NAME	VARCHAR2(40)
EMAIL	VARCHAR2(40)
PASSWORD	VARCHAR2(40)
NODE_ID	NUMBER(6,0)

MESSURMENT

Column Name	Data Type
NODE_ID	NUMBER(6,0)
MESSURMENT_ID	NUMBER(6,0)
SOIL_MOISTURE	NUMBER
AIR_TEMPERATURE	NUMBER
HUMIDITY	NUMBER
BRIGHTNESS	NUMBER
LIGHT_STATE	NUMBER

NODES

Column Name	Data Type
USER_ID	NUMBER(6,0)
PLANT_SPECIES_ID	NUMBER(6,0)
NODE_ID	NUMBER(6,0)
ASSIGNED_DATE	DATE
MAC_ADDRESS_	VARCHAR2(40)
NAME	VARCHAR2(40)

TASKS

Column Name	Data Type
NODE_ID	NUMBER(6,0)
TASKS_ID	NUMBER(6,0)
TASK_TYPE	VARCHAR2(40)
QUANTITY	NUMBER(3,0)
DUE_DATE	DATE
STATUS	VARCHAR2(40)
DATE-	DATE

PLANT_SPECIES

Column Name	Data Type
PLANT_SPECIES_ID	NUMBER(6,0)
BOTANIC_NAME	VARCHAR2(40)
SOIL_MOISTURE	VARCHAR2(40)
AIR_TEMPERATURE	VARCHAR2(40)
HUMIDITY	VARCHAR2(40)
BRIGHTNESS	VARCHAR2(40)
COMMON_NAME	VARCHAR2(40)
TIMESTAMP	TIMESTAMP(6)

- Each table must contain at least 5 records

TASKS-												+ v			
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	Sample Queries			
Query Count Rows Insert Row Load Data															
EDIT		NODE_ID		TASKS_ID		TASK_TYPE		QUANTITY		DUE_DATE		STATUS		DATE-	
		1002		102		watering		100		13-Jan-2025		done		01-Jan-2025	
		1001		101		lighting		230		01-Jan-2025		not done		19-Dec-2024	
		1003		103		watering		10		30-Jan-2025		not done		01-Jan-2025	
		1004		104		lighting		20		20-Jun-2006		done		19-Jun-2006	
		1005		105		watering		450		06-Dec-2024		done		01-Dec-2024	

USER-												+ v
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	Sample Queries
Query	Count Rows	Insert Row	Load Data									
EDIT		USER_ID		FIRST_NAME		LAST_NAME		EMAIL		PASSWORD		NODE_ID
	14	Maxime		Hell		Ma@14		14		1004		
	13	Marie		hell		M@13		13		1003		
	15	Ly		Nils		Ly@15		15		1005		
	11	Laura		Glode		l@11		11		1001		
	12	Patrice		helen		P@12		12		1002		

MESSURMENT													+ v
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	Sample Queries	
Query	Count Rows	Insert Row	Load Data										
EDIT	NODE_ID	MESSURMENT_ID		SOIL_MOISTURE		AIR_TEMPERATURE		HUMIDITY		BRIGHTNESS		LIGHT_STATE	
	1002	22	61	23	11	12	22						
	1004	24	55	11	23	9	10						
	1005	25	55	25	20	30	40						
	1003	23	60	18	33	13	11						
	1001	21	60	21	10	10	21						

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PLANT_SPECIES													+ v
Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	Sample Queries	
Query	Count Rows	Insert Row	Load Data										
EDIT	PLANT_SPECIES_ID	BOTANIC_NAME	SOIL_MOISTURE	AIR_TEMPERATURE	HUMIDITY	BRIGHTNESS	COMMON_NAME	TIMESTAMP					
	2	-	12	11	22	12	Lily	17-JAN-25 09:30.00.000000 AM					
	3	Mint	22	23	23	33	Mint	17-JAN-25 09:20.00.000000 AM					
	1	cactus	60	22	10	20	cactus	17-JAN-25 09:20.00.000000 AM					
	4	Hibiscus	32	22	12	22	Hibiscus	17-JAN-25 10:20.00.000000 PM					
	5	Henna	22	32	22	11	Henna	17-JAN-25 09:20.00.000000 PM					